

AUTONOID RACING

Complete Vehicle Integration Checklist: A Patent-Safe Blueprint for
Autonomous Off-Road Supremacy

PHASE 1: THE VISION

Avoiding prior art through novel physical implementation and heuristic-driven logic.

THE CORE TAKEAWAY



Novel Execution

Focus on physical mechanisms rather than algorithms. Invent new ways for AI to interact with the chassis.



Patent Safety

Bypass textbook SLAM and PID control. Use proprioceptive fusion to stay outside existing IP danger zones.



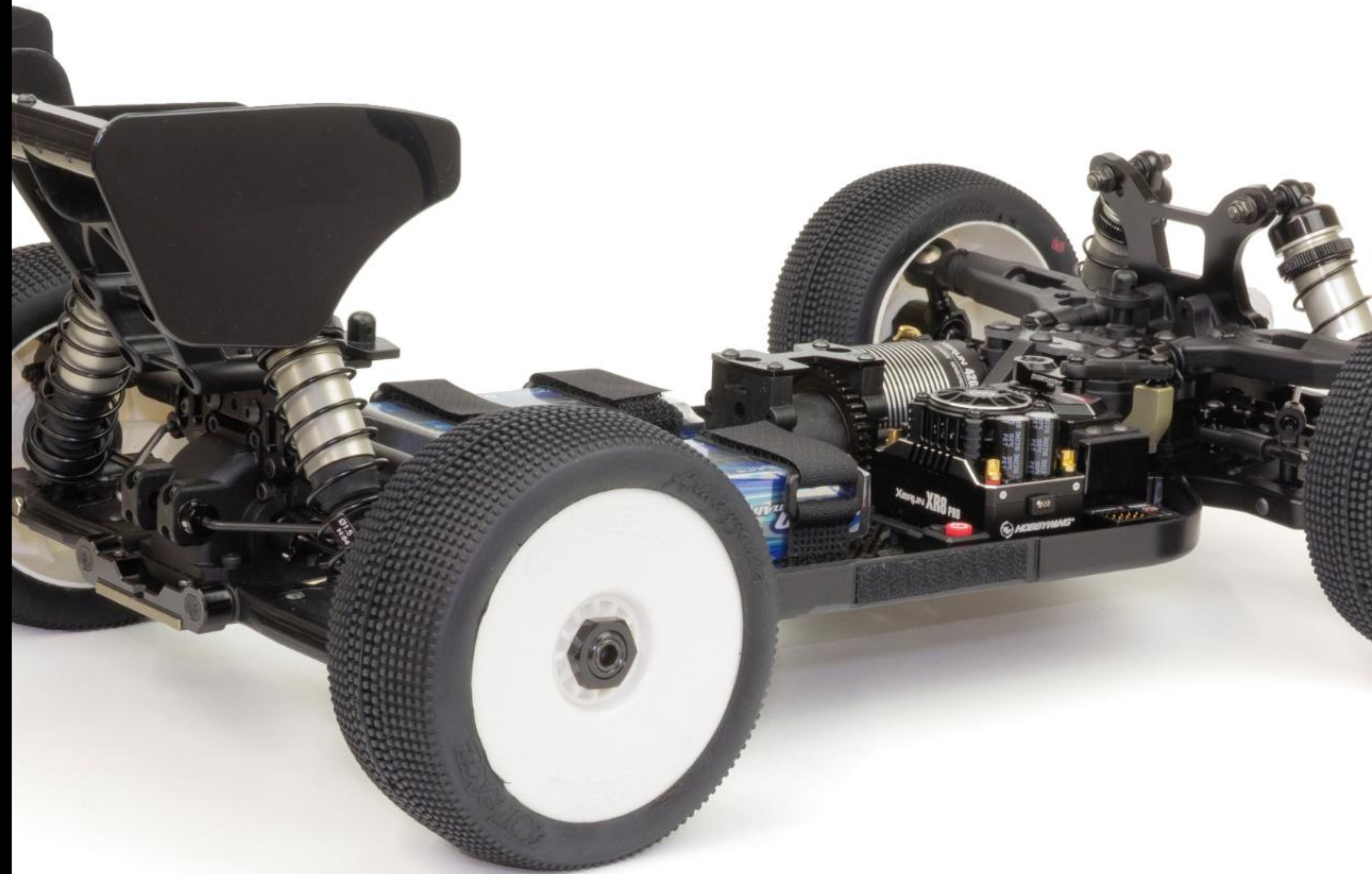
Proprioception

Prioritize internal state awareness via wheel-hub IMUs and ground-truth optical flow arrays.

CHASSIS FOUNDATION

The build begins with a high-performance 1/8 scale electric off-road buggy chassis.

- ✓ Reinforced cross-members
- ✓ Long-travel oil-filled shocks
- ✓ 4WD Brushless drivetrain



DRIVE-BY-WIRE & POWER

Actuation Layer

High-torque steering servos paired with MCU-controlled PWM for throttle and mechanical braking. Includes redundant remote kill-switch integration.

Power Rails

Isolated power systems: 4S-6S LiPo for drive motors, separate low-noise DC-DC regulated rails (12V/5V) for compute and sensor arrays.

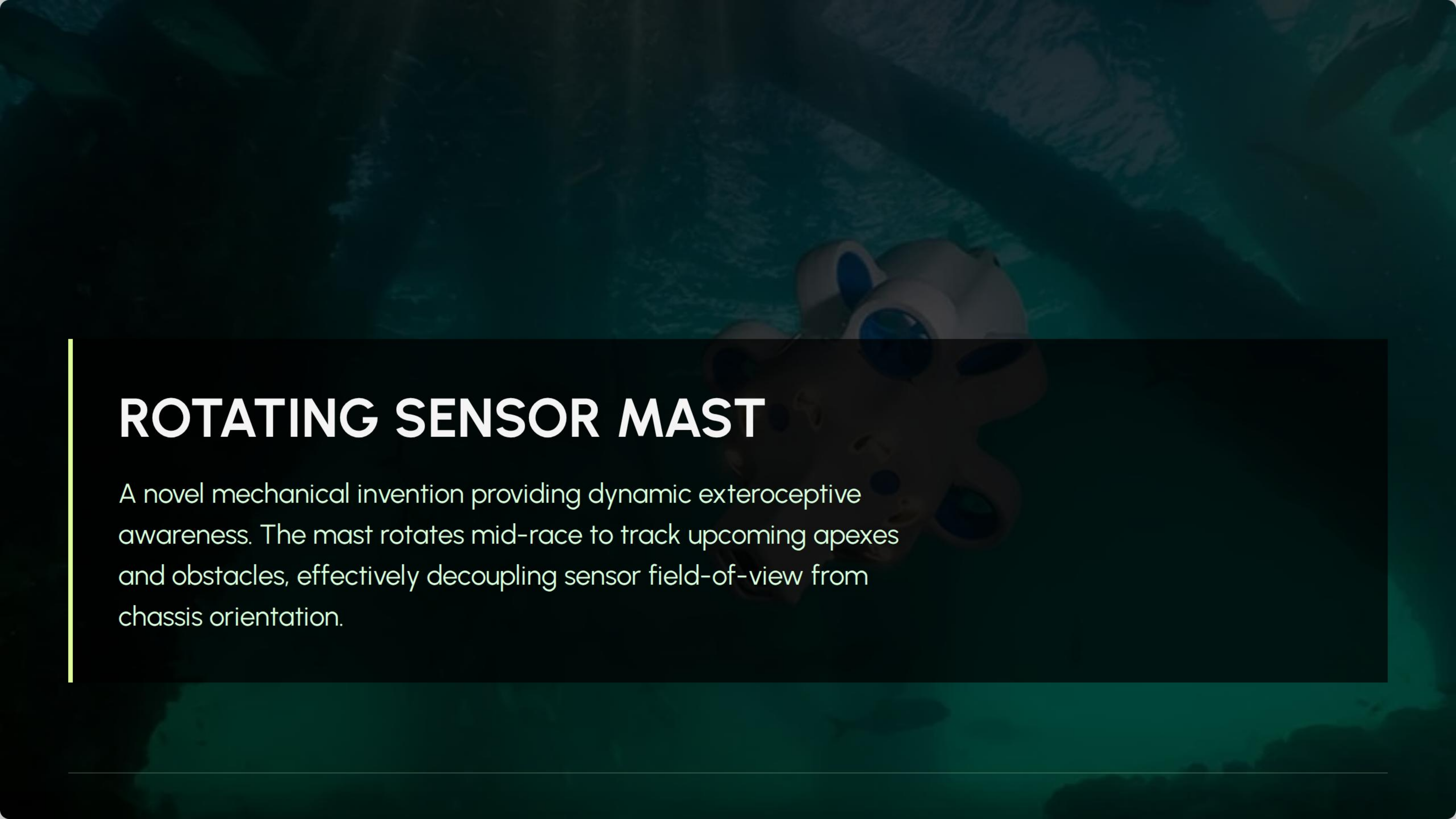
THE NEURAL CENTER

Jetson Orin Nano + MCU

The "Brain" utilizes an NVIDIA Jetson Orin Nano for high-speed vision and heuristic logic, while an STM32/ESP32 MCU handles deterministic real-time tasks like servo timing and sensor polling.

Integrated on a vibration-damped sandwich mount.





ROTATING SENSOR MAST

A novel mechanical invention providing dynamic exteroceptive awareness. The mast rotates mid-race to track upcoming apexes and obstacles, effectively decoupling sensor field-of-view from chassis orientation.

SENSOR ARRAY MAPPING

Location	Sensor Type	Primary Function
Rotating Mast	Diagonal Stereo Vision	Apex tracking and obstacle depth
Wheel Hubs	IMU Clusters	Localized vibration and friction detection
Under-Chassis	Optical Flow Array	True ground-speed & drift angle data
Chassis Center	6-Axis IMU + GNSS	Global pose and motion stabilization

BEHAVIORAL GRADES

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Proprioceptive Symmetry

A-

Rhythm Adherence

Dynamic Scoring Metrics

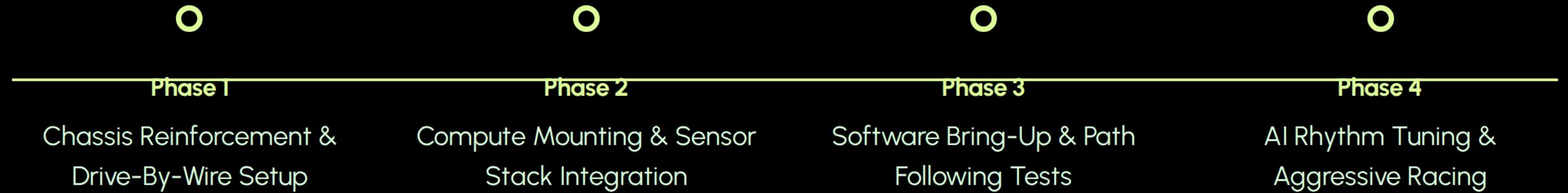
The Autonoid performance is graded on its ability to "feel" the track. Unlike standard metrics, we prioritize:

- **Corner Aggression Index:** Attack velocity vs Apex stability.
- **AI Bravery Score:** Risk-reward throttle ratio in off-road jumps.
- **Consistency Score:** Lap-to-lap rhythm variance.

SAFETY & HEARTBEAT

- ✓ **Hardware Kill Switch:** High-current relay cutting power to drive motors instantly via physical and remote triggers.
 - ✓ **Software Watchdog:** Heartbeat signal between Jetson and MCU. Any interruption triggers an "Autonomous Safe State."
 - ✓ **Safe State Protocol:** Microcontroller overrides commands to center steering and apply maximum motor braking.
 - ✓ **Vibration Watch:** Real-time monitoring of chassis twist to prevent mechanical fatigue failure during high-G maneuvers.
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BUILD TIMELINE



RACE READY

The Autonoid vehicle is now fully integrated and calibrated.



Questions? Let's take it to the track.

IMAGE SOURCES



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